

Supplementary Materials - Understanding Income Poverty in Mexico: A Comparative Framework Combining Statistical and Machine Learning Methods

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A Supplementary Materials

A.1 Modeling Framework

Logistic Regression under Complex Survey Data Logistic regression models must be adapted when applied to complex survey data to properly account for stratification, clustering, and unequal sampling probabilities. Let the population be stratified into H strata ($h = 1, \dots, H$), each containing several primary sampling units (PSUs, indexed by j), which in turn may include secondary sampling units (SSUs, indexed by i), and possibly individuals or observations within them (indexed by k). Thus, each observation is denoted by Y_{hjik} , with associated covariates $\mathbf{x}_{hjik}$ and sampling weights $w_{hjik} = 1/\pi_{hjik}$, where π_{hjik} is the inclusion probability [1]. The model is expressed as:

$$\text{logit} \{P(Y_{hjik} = 1 \mid \mathbf{x}_{hjik})\} = \mathbf{x}_{hjik}'\boldsymbol{\beta}, \quad (1)$$

where $\boldsymbol{\beta}$ are the regression coefficients. Estimation is done via maximum pseudo-likelihood, incorporating sampling weights into the log-likelihood function:

$$l_p(\boldsymbol{\beta}) = \sum w_{hjik} [y_{hjik} \ln P(Y = 1) + (1 - y_{hjik}) \ln(1 - P(Y = 1))]. \quad (2)$$

Machine and Deep Learning Models Models including Decision Trees (DT), Random Forests (RF), Logistic Regression (LR), Support Vector Classifier (SVC), Artificial Neural Networks (ANN), Extremely Randomized Trees (ERT), and a one-dimensional Convolutional Neural Network (1D-CNN) were employed. DT models classify data by recursively splitting it based on variables that best separate the target outcome. Their rule-based structure makes them easy to interpret and particularly suitable for identifying threshold-based relationships among socioeconomic variables; however, they tend to overfit if not properly pruned [2]. RF mitigates this limitation by aggregating multiple trees built on random subsets of data and predictors, enhancing generalization and

capturing heterogeneous poverty profiles through variable interactions [3]. LR models the probability of a binary outcome as a function of predictors and remains an essential benchmark due to its interpretability, especially when evaluating the directional effect of specific socioeconomic or demographic attributes [3]. SVC identifies an optimal separating hyperplane between poverty classes and, by incorporating nonlinear kernels, can model complex boundaries that arise from intertwined social and economic indicators [4]. ANN leverages interconnected hidden layers to represent nonlinear associations, offering flexibility to model latent behavioral or contextual dependencies that linear methods might miss [3]. ERT builds multiple randomized decision trees where both feature selection and split thresholds are stochastically determined, improving variance reduction and robustness [5].

Finally, 1D-CNNs apply convolutional filters to structured input vectors, enabling the extraction of local and hierarchical feature patterns. In the context of poverty classification, this allows the network to capture subtle dependencies between consecutive or related socioeconomic indicators—such as income, education, and housing quality—without requiring explicit feature engineering. This property makes CNNs particularly well-suited for high-dimensional, ordered data representations, where latent correlations play a role in predicting poverty outcomes [3].

B Supplementary Tables

Demographic variables (age, total_residents, children) capture household composition and dependency. Sociocultural and human-capital indicators (sex, literacy, hli, household_head_edu) reflect structural inequalities in gender, education, and ethnicity. Access-related variables (public_healthcare, medical_attention, electricity, water_drainage, connectivity, basic_energy_equipment) represent material well-being and service provision [7]. Economic inclusion is proxied by bank_card and housing_tenure, while urban_rural and region account for spatial heterogeneity. Household_occupation captures economic activity, and consumption expenditure proxies current welfare [6].

Table S1: Operational definitions and characteristics of study variables.

Variable	Alias	Values	Definition
Age	age	–	Age of the household head.
Household size	total_residents	–	Total number of members in the household.
Number of children	children	–	Number of household members under 18 years old.
Consumption Expenditure	consumption expenditure	–	Logarithm of the household’s total consumption expenditure ^a .
Gender	sex	Male (REF) Female	Gender of the household head.
Food insecurity	food_insecurity	Secure (REF) Mild insecurity Moderate insecurity Severe insecurity	Level of food insecurity based on standardized assessment.
Area of residence	urban_rural	Rural (REF) Urban	Area of the household’s location.

Table S1: *Cont.*

Variable	Alias	Values	Definition
Literacy	literacy	No (REF) Yes	Whether the household head can read and write.
Indigenous language	hli	No (REF) Yes	Whether the household head speaks an indigenous language.
Health Insurance	public_healthcare	Yes (REF) No	Whether household is covered by the health insurance <i>Seguro Popular</i> .
Medical attention	medical_attention	Yes (REF) No	Whether the household is affiliated to receive medical care from an institution.
Bank card	bank_card	No (REF) Yes	Whether the household head has access to a bank or commercial card.
Electricity	electricity	Yes (REF) No	Whether the household has access to electricity.
Connectivity	connectivity	No (REF) Yes	Whether the household has access to communication services ^b .
Water and sewage	water_drainage	No (REF) Yes	Access to both piped water and sewage services at the household.
Child labor	child_labor	No (REF) Yes	Whether children in the household engage in labor.
Basic energy equipment	basic_energy_equipment	No (REF) Yes	Access to at least one heating or cooling device at the household.
Housing tenure	housing_tenure	No (REF) Yes	Whether the household head holds legal ownership of the dwelling.
Education level	household_head_edu	No schooling (REF) Primary Middle School High school Higher education	Highest level of education attained by the household head.
Region	region	Center-North (REF) Northwest Southeast Northeast West Southwest Center-South East	Geographic region in which the household is located.
Main household Occupation	household_occupation	Industrial (REF) Commercial Services Agriculture Livestock Gathering Logging Hunting Fishing More than one	Main economic activity in the household ^c .

(a) Based on food and transportation expenses; (b) Access to at least one service, such as telephone, mobile phone, television, or internet; (c) Based on the predominant economic activity considering all household members.

Table S2 outlines the tested hyperparameters based on grid search with 10-fold cross-validation.

Table S2: Definition of hyperparameters for ML models.

Algorithm	Hyperparameter	Definition
LR	–	Not applicable; no tunable predictive hyperparameters.
ANN	Hidden layer sizes	Number of neurons in the hidden layer(s) of the network.
	Activation	Activation function used in the hidden layers.
	Solver	Optimization algorithm used for weight updating.
	Initial learning rate	Starting step size for updating weights during training.
	Learning rate	Strategy for adapting the learning rate.
RF	Momentum	Factor that accelerates SGD in the relevant direction.
	Max iterations	Maximum number of training epochs.
	Number of estimators	Number of trees in the forest.
	Max depth	Maximum depth of each tree.
	Max features	Number of features when splitting a node.
DT	Criterion	Function to measure the quality of a split.
	Max depth	Maximum depth of the tree.
	Max features	Number of features when splitting a node.
	Criterion	Function to measure the quality of a split.
	Criterion	Function to measure the quality of a split.
ERT	Number of estimators	Total number of trees in the ensemble.
	Maximum depth	Maximum level of tree growth.
	Maximum features	Features evaluated at each split.
	Minimum samples split	Minimum samples to split a node.
	Minimum samples leaf	Minimum samples per leaf node.
SVC	Bootstrap	Whether sampling with replacement is used.
	Criterion	Metric for evaluating split quality.
	Regularization parameter (C)	Balances margin width and classification errors.
	Kernel coefficient (γ)	Controls influence of training samples.
	Kernel type	Defines the transformation function.

The 1D-CNN was developed in two phases: first, architectural tuning, including the number of filters, kernel size, and hidden layers; then, training parameter adjustment, such as epochs and batch size. The initial baseline architecture included convolutional, pooling, and dense layers (Table S3), which were progressively refined for improved performance and generalization.

Table S3: 1D-CNN architecture layers (architectural tuning phase).

Layer Name
Conv1D
Conv1D
MaxPooling1D
Conv1D
GlobalAveragePooling1D
Dense
Dense

Moreover, Table S4 presents the initial parameters used in the baseline model.

Table S4: Base parameters for the 1D-CNN model.

Parameter	Value
Optimizer	Adam
Batch size	32
Kernel size	2
Epochs	200
Loss function	Binary Cross-Entropy
Activation function	ReLU
Activation function	Sigmoid
Learning rate	0.001

CNN tuning involved iterative 10-fold cross-validation on the baseline structure. Filter values and kernel sizes (2–11) were varied to balance pattern recognition and computational efficiency. Architectural enhancements such as batch normalization, dropout, flattening, and average pooling were also introduced.

Training parameters were also optimized: epoch ranges (25–200) and batch sizes (32–160) were tested to balance convergence speed and generalization. Smaller batches tended to improve generalization, while larger ones affected training stability. The final CNN architecture reflects the result of successive refinements across structure and training to ensure classification performance.

Moreover, performance was assessed based on the confusion matrix, as detailed in Table S5.

Table S5: Confusion matrix for models' evaluation.

		Predicted Class	
True Class		Negative	Positive
Negative	True Negative (<i>TN</i>)		False Positive (<i>FP</i>)
Positive	False Negative (<i>FN</i>)		True Positive (<i>TP</i>)

Conventional classification metrics used in the study are formulated below:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (5)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

$$\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (7)$$

The univariate analysis indicates differences between income-poor and non-poor households across a range of demographic and socioeconomic characteristics, as illustrated in Tables S6 and S7, respectively.

Table S6: Descriptive statistics for continuous variables by income poverty status.

Variable	Income Poverty Status		
	Yes (n = 26,225)	No (n = 65,189)	t-test
	Mean (S.D.)	Mean (S.D.)	p-value
Age	51.10 (16.10)	51.70 (15.90)	<0.01 ***
Household size	4.10 (2.00)	3.20 (1.70)	<0.01 ***
Number of children	1.50 (1.40)	0.80 (1.00)	<0.01 ***
Consumption expenditure	8.20 (0.60)	8.60 (0.70)	<0.01 ***

Notes.- S.D.: Standard Deviation; Significance levels: *** $p < 0.01$.

Table S7: Descriptive statistics for categorical variables by income poverty status.

		Income Poverty Status		χ^2 Test
		Yes (n = 26,225)	No (n = 65,189)	
		n (%)	n (%)	p-value
Area of residence	Urban	14,162 (54.0%)	41,427 (63.5%)	<0.01 ***
	Rural	12,063 (46.0%)	23,762 (36.5%)	
Gender	Male	18,147 (69.2%)	43,705 (67.0%)	<0.01 ***
	Female	8,078 (30.8%)	21,484 (33.0%)	
Literacy	Yes	23,279 (88.8%)	62,200 (95.4%)	<0.01 ***
	No	2,946 (11.2%)	2,989 (4.6%)	
Employment in main occupation	Yes	19,380 (73.9%)	48,964 (75.1%)	0.714 [N.S.]
	No	6,845 (26.1%)	16,225 (24.9%)	
Food insecurity	Secure	15,484 (59.0%)	52,295 (80.2%)	<0.01 ***
	Mild	5,116 (19.5%)	7,030 (10.8%)	
	Moderate	3,142 (12.0%)	3,336 (5.1%)	
	Severe	2,483 (9.5%)	2,528 (3.9%)	
Indigenous language	Yes	4,292 (16.4%)	3,681 (5.6%)	<0.01 ***
	No	21,933 (83.6%)	61,508 (94.4%)	
Health insurance	Yes	15,965 (60.9%)	21,464 (32.9%)	<0.01 ***
	No	10,260 (39.1%)	43,725 (67.1%)	
Medical attention	Yes	5,200 (19.8%)	15,941 (24.5%)	<0.01 ***
	No	21,025 (80.2%)	49,248 (75.5%)	
Bank card	Yes	3,826 (14.6%)	23,757 (36.4%)	<0.01 ***
	No	22,399 (85.4%)	41,432 (63.6%)	
Basic energy equipment	Yes	25,864 (98.6%)	61,434 (94.2%)	<0.01 ***
	No	361 (1.4%)	3,755 (5.8%)	
Water and sewage	Yes	13,160 (50.2%)	42,698 (65.5%)	<0.01 ***
	No	13,065 (49.8%)	22,491 (34.5%)	
Electricity	Yes	25,571 (97.5%)	64,289 (98.6%)	<0.01 ***
	No	654 (2.5%)	900 (1.4%)	
Housing tenure	Yes	18,373 (70.1%)	40,367 (61.9%)	<0.01 ***
	No	7,852 (29.9%)	24,822 (38.1%)	
Child labor	Yes	2,414 (9.2%)	3,478 (5.3%)	<0.01 ***
	No	23,811 (90.8%)	61,711 (94.7%)	
Connectivity	Yes	24,896 (94.9%)	55,921 (85.8%)	<0.01 ***
	No	1,329 (5.1%)	9,268 (14.2%)	
Education level	No education	2,530 (9.6%)	2,913 (4.5%)	<0.01 ***
	Primary	10,417 (39.7%)	17,185 (26.4%)	
	Middle school	8,199 (31.3%)	17,638 (27.1%)	
	High school	3,637 (13.9%)	12,659 (19.4%)	
	Higher education	1,442 (5.5%)	14,794 (22.7%)	
Region	Central-North	4,154 (15.8%)	10,828 (16.6%)	<0.01 ***
	Northwest	3,754 (14.3%)	16,680 (25.6%)	
	Southeast	2,978 (11.4%)	6,776 (10.4%)	
	Northeast	2,182 (8.3%)	8,065 (12.4%)	
	West	2,292 (8.7%)	7,917 (12.1%)	
	Southwest	4,098 (15.6%)	3,257 (5.0%)	
	Central-South	2,629 (10.0%)	6,078 (9.3%)	
	East	4,138 (15.8%)	5,588 (8.6%)	
Main household occupation	Industrial	97 (0.4%)	1,698 (2.6%)	<0.01 ***
	Commercial	912 (3.5%)	7,397 (11.3%)	
	Services	350 (1.3%)	1,742 (2.7%)	
	Agriculture	1,837 (7.0%)	4,347 (6.7%)	
	Livestock	936 (3.6%)	2,850 (4.4%)	
	Gathering	2,818 (10.7%)	3,235 (5.0%)	
	Logging	1,575 (6.0%)	2,859 (4.4%)	
	Hunting	1,268 (4.8%)	4,354 (6.7%)	
	Fishing	6,775 (25.8%)	9,359 (14.4%)	
	Multiple	9,657 (36.8%)	27,348 (42.0%)	

Notes.- Significance levels: *** $p < 0.01$. [N.S.] non-significant.

Based on the initial 1D-CNN architecture in Table S3, filter configurations and results for the proposed models are shown in Table S8.

Table S8: Proposed models for assessment of filter values variations with performance metrics.

Model	Filter Values	Accuracy (%) ($\pm$ Std. Dev.)
Model 1	Filter: 75-50-25 (Conv1D) & 25-1 (Dense)	79.51 ($\pm$ 0.21)
Model 2	Filter: 50-25-10 (Conv1D) & 10-1 (Dense)	79.73 ($\pm$ 0.23)
Model 3	Filter: 30-15-5 (Conv1D) & 10-1 (Dense)	79.19 ($\pm$ 0.23)
Model 4	Filter: 20-10-5 (Conv1D) & 5-1 (Dense)	78.72 ($\pm$ 0.29)
Model 5	Filter: 100-50-25 (Conv1D) & 20-1 (Dense)	79.50 ($\pm$ 0.22)
Model 6	Filter: 120-100-120 (Conv1D) & 40-1 (Dense)	79.53 ($\pm$ 0.10)
Model 7	Filter: 150-100-50 (Conv1D) & 25-1 (Dense)	79.56 ($\pm$ 0.09)
Model 8	Filter: 20-10-5 (Conv1D) & 10-1 (Dense)	78.50 ($\pm$ 0.37)

Notes.- Bold values indicate the best model based on the performance.

Model 2 achieved the highest average accuracy (79.73%), indicating stable performance by enabling effective feature extraction and generalization. In contrast, Model 8 obtained the lowest accuracy (78.50%), likely due to its limited filter configuration. Hence, Model 2 was selected as the 1D-CNN architecture.

As a next step, Table S9 presents the tested kernel sizes and their results.

Table S9: Evaluation accuracy of 1D-CNN with different kernel sizes.

Kernel Size	2	3	5	7	11
Avg. acc. (%)	79.08	79.27	79.39	79.62	79.39
Std. dev.	$\pm$ 0.20	$\pm$ 0.19	$\pm$ 0.17	$\pm$ 0.22	$\pm$ 0.16

Notes.- Bold values indicate the best kernel value based on the performance.

A kernel size of 7 achieved the highest average accuracy (79.62%), indicating effective feature extraction and consistent performance. In comparison, kernel size 2 produced lower accuracy (79.08%), suggesting instability across iterations. Based on these results, kernel size 7 was selected.

Table S10 summarizes different proposed model variations.

Table S10: Operational definition of the proposed architectures with updated hyper-parameters and performance metrics.

Architecture	Description	Hyper-Parameters Values	Accuracy (%) ($\pm$ Std. Dev.)
Architecture 1 (A1)	2 convolutional layers followed by a max-pooling layer, another convolutional layer, global average pooling layer, and 2 dense layers	Filters: 50-25-10 (conv), 10-1 (dense). Kernel: 7.	79.50 ($\pm$ 0.24)
Architecture 2 (A2)	4 additional convolutional layers based on A1	Filters: 50-25-50-25-10 (conv), 10-1 (dense). Kernel: 7.	79.36 ($\pm$ 0.13)
Architecture 3 (A3)	2 convolutional layers, max-pooling, another conv layer, global average pooling, and 4 consecutive dense layers	Filters: 50-25-50 (conv), 30-20-10-1 (dense). Kernel: 7.	79.35 ($\pm$ 0.13)
Architecture 4 (A4)	4 convolutional layers, max-pooling, another conv layer, global average pooling, and 4 consecutive dense layers	Filters: 50-25-50-50-25 (conv), 30-20-10-1 (dense). Kernel: 7.	79.42 ($\pm$ 0.18)
Architecture 5 (A5)	A dropout layer added after the first dense layer in A1	Filters: 50-25-10 (conv), 10-1 (dense). Kernel: 7. Dropout: 0.1	79.49 ($\pm$ 0.15)
Architecture 6 (A6)	Dropout increased to 0.2 from A5	Filters: 50-25-10 (conv), 10-1 (dense). Kernel: 7. Dropout: 0.2	79.41 ($\pm$ 0.15)
Architecture 7 (A7)	Batch normalization added after each convolutional and dense layer in A6	Filters: 50-25-10 (conv), 10-1 (dense). Kernel: 7. Dropout: 0.1.	79.98 ($\pm$ 0.07)
Architecture 8 (A8)	Flatten layer replaces global average pooling in A7	Filters: 50-25-10 (conv), 10-1 (dense). Kernel: 7. Dropout: 0.1.	80.17 ($\pm$ 0.13)
Architecture 9 (A9)	Max-pooling replaced with average pooling in A7	Filters: 50-25-10 (conv), 10-1 (dense). Kernel: 7. Dropout: 0.1.	80.02 ($\pm$ 0.06)
Architecture 10 (A10)	Flatten layer replaces global average pooling and max-pooling replaced with average pooling in A7	Filters: 50-25-10 (conv), 10-1 (dense). Kernel: 7. Dropout: 0.1.	80.08 ($\pm$ 0.07)

Notes.- Bold values indicate the best architecture based on the performance.

Performance differences across architectures confirm that structural design influences performance. Architecture A8 achieved the highest average accuracy (80.17%), indicating the effectiveness of batch normalization and the placement of a flatten layer instead of a global average pooling layer in enhancing both performance and stability. In contrast, Architecture A3, which introduced additional dense layers, resulted in the lowest accuracy (79.35%), suggesting that increased complexity may hinder consistent learning. Based on these findings, Architecture A8 was selected as the optimal structure.

Table S11 summarizes the results obtained from varying two key training parameters: the number of epochs and the batch size.

Table S11: Performance comparison for epochs and batch sizes in model tuning.

	Epochs					Batch Size				
	25	50	100	150	200	32	64	96	128	160
Avg. acc. (%)	80.03	79.95	79.94	79.94	79.99	80.04	80.08	80.10	80.22	80.07
Std. dev.	$\pm$ 0.06	$\pm$ 0.03	$\pm$ 0.07	$\pm$ 0.04	$\pm$ 0.04	$\pm$ 0.05	$\pm$ 0.05	$\pm$ 0.04	$\pm$ 0.09	$\pm$ 0.29

Notes.- Bold values represent the highest accuracy obtained for each parameter.

Training with 25 epochs yielded the highest average accuracy, indicating efficient learning and consistent performance. Longer training, such as 200 epochs, suggests potential overfitting. Therefore, a value of 25 epochs was selected. Furthermore, a batch size of 128 achieved the highest average accuracy, indicating effective learning dynamics and consistent performance. In contrast, batch size 32 produced the lowest accuracy, suggesting less stable training. Based on these results, batch size 96 was selected as the optimal value.

C Supplementary Figures

The methodological framework followed in this study is illustrated in Figure S8, which outlines the main stages of data preparation and model development.

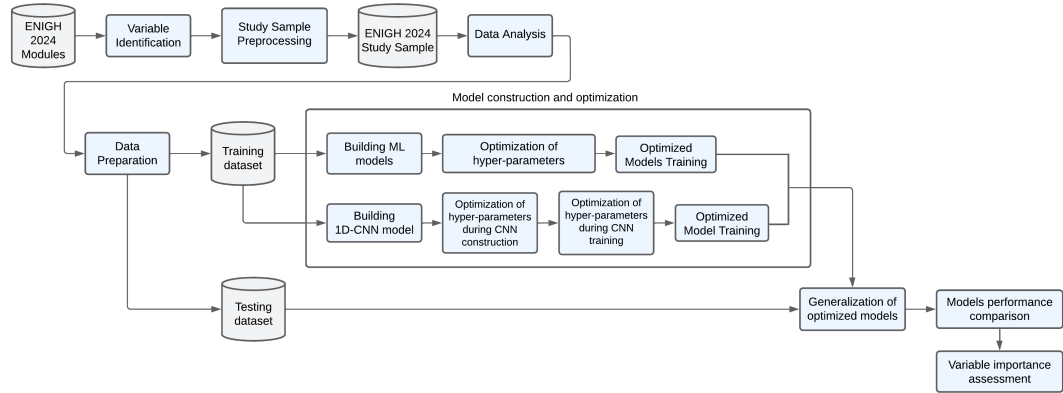


Fig. S1: Methodological framework of the study.

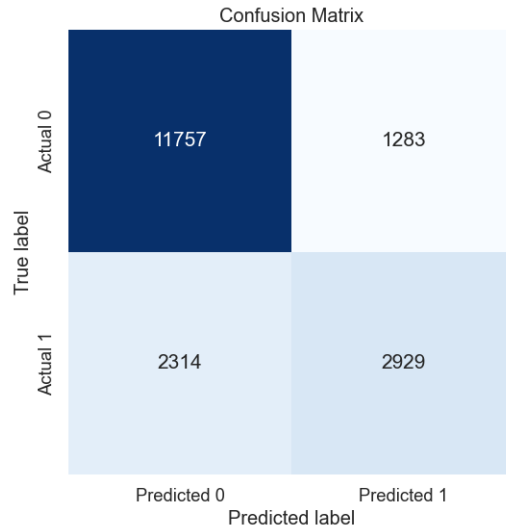


Fig. S2: Confusion matrix for the 1D-CNN model evaluated on the testing set.

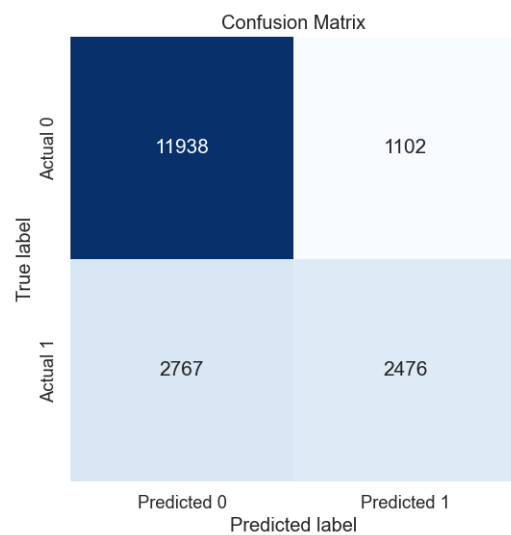


Fig. S3: Confusion matrix for the logistic regression model evaluated on the testing set.

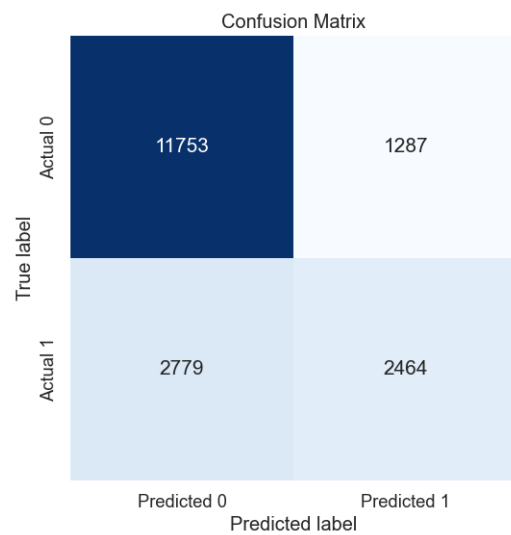


Fig. S4: Confusion matrix for the DT model evaluated on the testing set.

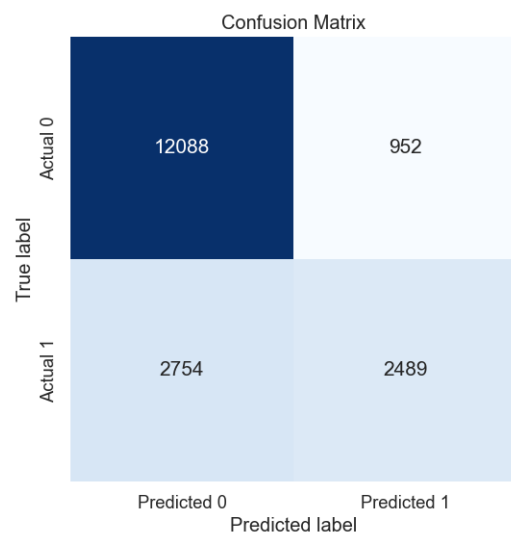


Fig. S5: Confusion matrix for the RF model evaluated on the testing set.

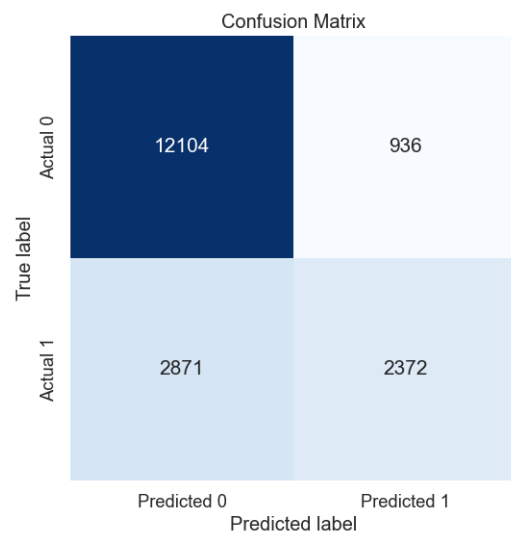


Fig. S6: Confusion matrix for the ERT model evaluated on the testing set.

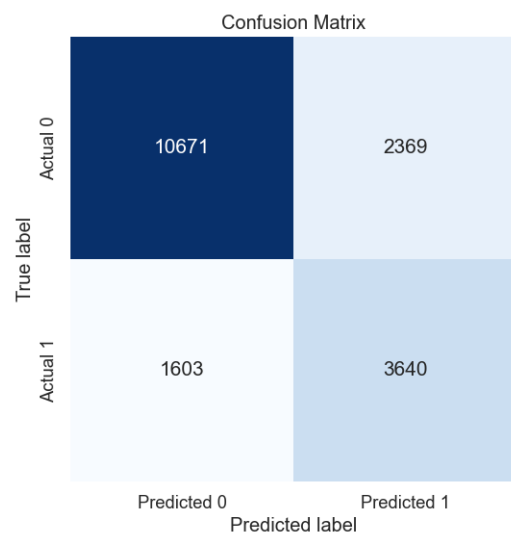


Fig. S7: Confusion matrix for the ANN model evaluated on the testing set.

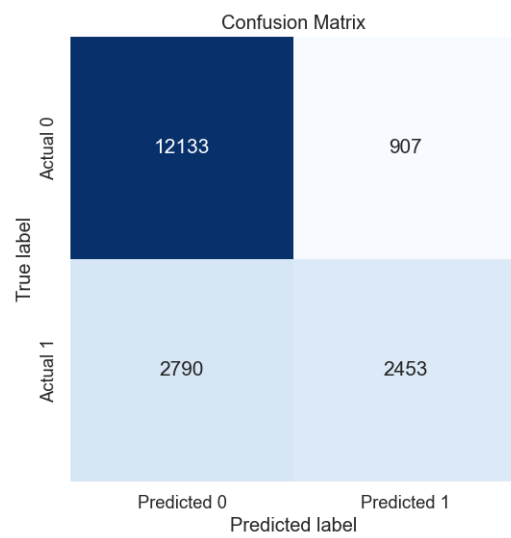


Fig. S8: Confusion matrix for the SVC model evaluated on the testing set.

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